

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0378

Lammers 138 kV
September 03, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0378. The Interconnection Customer proposes a 173.5 MW battery energy storage facility interconnecting at the Lammers 138 kV substation in the Maryland area. The request was received on September 10, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 171 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$50,843,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Brainerd - Memphis Jct 138.0 kV	Memphis Junction - Persimmon Gap 138.0 kV	145.5	227	35.7
Brainerd - Persimmon Gap 138.0 kV	Persimmon Gap Substation - Memphis Junction 138.0 kV Line	141.4	242	35.7
Sollers Warf - Calvert Cliffs 138.0 kV Ckt 2	Sollers Warf - Calvert Cliffs 138.0 kV Cir. 1	138.7	321	34.9
Memphis Jct - Persimmon Gap 138.0 kV	Persimmon Gap - Brainerd 138.0 kV	137.7	259	37.3
Calvert Cliffs - Prtt 2 138.0 kV	Calvert Cliffs 345 kV/138 kV Transformer Bank	136.9	219	22.8
Calvert Cliffs - Sollers Warf 138.0 kV	Calvert Cliffs - Sollers Warf 138.0 kV Circuit 2	136.5	326	34.9
Bozman - Spencer 138.0 kV	Saddlebrooke Ranch - Lammers 138.0 kV	132.8	183	23.9
Memphis Junction - Lammers 138.0 kV	Vienna Operations - Easton 2 345.0 kV	122.5	354	46.8
Greenstone - Boman 138.0 kV	Sollers Warf - Persimmon Gap 138.0 kV	121.6	204	15.8
Sollers Warf Substation - Persimmon Gap 138.0 kV	Vienna Operations - Easton 2 345.0 kV	117.7	404	46.8
Easton Substation - Navy Mole 138.0 kV line	Saddlebrooke Ranch - Lammers 138.0 kV L	117.5	322	21.4
Hansford - Sooner Substation 138.0 kV line	Sunset Substation - Saddlebrooke Ranch 138.0 kV line	115.4	253	33.9
Dobbins - Easton 138.0 kV	Lammers - Saddlebrooke Ranch 138.0 kV	114.4	360	21.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dobbins - Spencer 138.0 kV	Lammers - Saddlebrooke Ranch Substation 138.0 kV Line	112.9	381	21.4
North Lebanon - Fairmont 138.0 kV Line 1	N Lebanon - Fairmont 138.0 kV	112.8	320	7.0
Saddlebrooke Ranch - Hansford 138.0 kV	Sunset - Saddlebrooke Ranch Substation 138.0 kV line	111.7	293	33.9
Chemolite - Lammers 138.0 kV	Sollers Warf - Persimmon Gap 138.0 kV	110.9	170	32.0
Sooner - Kiwi 138.0 kV	Sunset - Saddlebrooke Ranch 138.0 kV	110.1	199	27.6
Bond-Elk Grove Florin - Fairmont 138.0 kV	Bond-ElkGrove Florin - Fairmont 138.0 kV	109.9	322	7.0
Foreman Cement - Greentone 138.0 kV Line	Sollers Warf - Persimmon Gap 138.0 kV	108.6	236	15.8
Foreman Cement - Whispering Pines 138.0 kV	Sollers Warf - Persimmon Gap 138.0 kV	107.8	253	15.8
Chemolite Substation - Randall County Substation 138.0 kV	Sollers Warf - Persimmon Gap 138.0 kV	105.2	190	32.0
Bond-Elk Grove Florin - Fairmont 138.0 kV Line 2	Bond-Elk Grove Florin - Fairmont 138.0 kV #1	104.1	339	7.0
Hewitt Rd - Grahamsville 138.0 kV	Greening 345 kV/138 kV TX	102.0	386	7.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Brainerd - Memphis Jct 138.0 kV	Rebuild line with higher-capacity conductor	\$4,330,000	48% (\$2,062,000)
Persimmon Gap - Brainerd 138.0 kV	Rebuild line with higher-capacity conductor	\$3,866,000	48% (\$1,842,000)
Calvert Cliffs - Sollers Warf 138.0 kV Line 2	Rebuild line with higher-capacity conductor	\$1,058,000	31% (\$331,000)
Persimmon Gap - Memphis Jct 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,010,000	48% (\$1,434,000)
Calvert Cliffs - Pratt2 Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,817,000	49% (\$1,393,000)
Sollers Warf - Calvert Cliffs 138.0 kV Line	Rebuild line with higher-capacity conductor	\$928,000	31% (\$290,000)
Bozman - Spencer 138.0 kV	Rebuild line with higher-capacity conductor	\$16,864,000	50% (\$8,469,000)
Lammers - Memphis Jct 138.0 kV	Rebuild line with higher-capacity conductor	\$19,690,000	24% (\$4,692,000)
Greenstone - Boman 138.0 kV	Rebuild line with higher-capacity conductor	\$19,640,000	37% (\$7,225,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Persimmon Gap - Sollers Warf 138.0 kV	Rebuild line with higher-capacity conductor	\$18,690,000	24% (\$4,454,000)
Navy Mole - Easton 138.0 kV	Rebuild line with higher-capacity conductor	\$4,597,000	13% (\$590,000)
Hansford - Sooner 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,191,000	51% (\$1,625,000)
Dobbins - Easton 138.0 kV	Rebuild line with higher-capacity conductor	\$2,271,000	13% (\$292,000)
Spencer - Dobbins 138.0 kV	Rebuild line with higher-capacity conductor	\$5,311,000	13% (\$682,000)
N Lebanon - Fairmont 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,614,000	3% (\$101,000)
Saddlebrooke Ranch - Hansford 138.0 kV	Rebuild line with higher-capacity conductor	\$6,789,000	51% (\$3,458,000)
Chemolite - Lammers 138.0 kV	Rebuild line with higher-capacity conductor	\$5,166,000	57% (\$2,933,000)
Sooner - Kiwi 138.0 kV	Rebuild line with higher-capacity conductor	\$4,924,000	48% (\$2,362,000)
Bond-Elk Grove Florin - Fairmont 138.0 kV	Reconductor line	\$694,000	3% (\$19,000)
Greenstone - Foreman Cement 138.0 kV	Reconductor line	\$2,496,000	37% (\$918,000)
ForemanCement - Whispering Pines 138.0 kV	Reconductor line	\$10,875,000	37% (\$4,001,000)
Chemolite Substation - Randall County 138.0 kV	Reconductor line	\$2,707,000	57% (\$1,537,000)
Bond-Elk Grove Florin - Fairmont 138.0 kV #2	Reconductor line	\$1,116,000	3% (\$31,000)
Hewitt Rd - Grahamsville 138.0 kV	Reconductor line	\$2,432,000	4% (\$102,000)
	Total allocated to project		\$50,843,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.